

A Constant Lower Bound for the Union-Closed Sets Conjecture

An entropy-method proof after Gilmer (2022)

1 Preliminaries

Throughout, $\log$ denotes the base-2 logarithm, so entropies are measured in bits. We work over a finite universe $[n] = \{1, \dots, n\}$ and identify a subset $A \subseteq [n]$ with its indicator vector $(A_1, \dots, A_n) \in \{0, 1\}^n$, where $A_i = 1$ iff $i \in A$. For a sequence we write $A_{<i} = (A_1, \dots, A_{i-1})$. For two bits we write $A_i \vee B_i := \max(A_i, B_i)$, so that the indicator of $A \cup B$ at coordinate i is $(A \cup B)_i = A_i \vee B_i$.

Throughout, the displayed numerical constants (0.01, 0.1, 1.26, 1.4, 1.62) are the explicit un-optimized values from Gilmer's argument and are not universal constants in the floating sense; we therefore do not invoke the universal-constant convention and track each value explicitly. The single bound constant we name, c , is fixed to the value 0.01 in Theorem 7.1.

Definition 1.1 (Union-closed family). A family $\mathcal{F} \subseteq 2^{[n]}$ is *union-closed* if $A \cup B \in \mathcal{F}$ for all $A, B \in \mathcal{F}$.

Definition 1.2 (Binary entropy). The *binary-entropy function* $h: [0, 1] \rightarrow [0, 1]$ is

$$h(p) := -p \log p - (1-p) \log(1-p),$$

with the convention $0 \log 0 = 0$, so $h(0) = h(1) = 0$. For a random variable Z on a finite set we write $H(Z)$ for its Shannon entropy, and $H(X \mid Y)$ for the conditional entropy. For a single bit X with $\mathbb{P}[X = 1] = p$ we have $H(X) = h(p)$.

We record the information-theoretic facts the proof uses; all are standard (see Cover and Thomas [1]).

Fact 1.3 (Information-theoretic toolkit, [1]). Let X, Y, Z and $X_1, \dots, X_n$ be random variables on finite sets. Then:

- (i) (*Maximum entropy.*) If Z takes values in a finite set $\mathcal{S}$ then $H(Z) \leq \log |\mathcal{S}|$, with equality iff Z is uniform on $\mathcal{S}$.
- (ii) (*Chain rule.*) $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i \mid X_{<i})$.
- (iii) (*Data processing.*) If $X \rightarrow Y \rightarrow Z$ is a Markov chain then $I(X; Z) \leq I(X; Y)$, where $I(\cdot; \cdot)$ is the mutual information.
- (iv) (*Concavity.*) The binary-entropy function h is concave on $[0, 1]$.

2 Conditional-entropy monotonicity under coarsening

The information-theoretic formulation of the proof rests on a single structural fact: conditioning on *more* information cannot increase conditional entropy. We state it in the form used at the crucial step of the main theorem.

Lemma 2.1 (Coarsening monotonicity). *Let X, Y be random variables on finite sets and let f be any (deterministic) function of Y . Then*

$$H(X | Y) \leq H(X | f(Y)).$$

Proof. The triple $X \rightarrow Y \rightarrow f(Y)$ is a Markov chain, since $f(Y)$ is a deterministic function of Y and hence conditionally independent of X given Y . We compute

$$\begin{aligned} I(X; f(Y)) &\stackrel{(a)}{\leq} I(X; Y) \\ H(X) - H(X | f(Y)) &\stackrel{(b)}{\leq} H(X) - H(X | Y) \\ H(X | Y) &\stackrel{(c)}{\leq} H(X | f(Y)), \end{aligned}$$

where step (a) is the data-processing inequality Item 1.3(iii) applied to the Markov chain $X \rightarrow Y \rightarrow f(Y)$, step (b) rewrites each mutual information through the identity $I(X; W) = H(X) - H(X | W)$, and step (c) cancels the common term $H(X)$ and rearranges. This gives the assertion. $\square$

3 Single-variable entropy inequalities

For two independent bits with success probabilities p, p' , the union (bitwise OR) succeeds with probability

$$\mathbb{P}[\text{bit} \vee \text{bit}' = 1] = 1 - (1 - p)(1 - p') = p + p' - pp'.$$

The two lemmas below control the binary entropy $h(p + p' - pp')$ of this OR in the two regimes the key lemma needs: both probabilities small (Lemma 3.1), and one of them arbitrary (Lemma 3.2). Both are consequences of concavity of h (Item 1.3(iv)).

Lemma 3.1 (Small-probability amplification). *For all $p, p' \in [0, 0.1]$,*

$$h(p + p' - pp') \geq 1.4 \cdot \frac{h(p) + h(p')}{2}.$$

Proof. The inequality holds with both sides 0 when $p = p' = 0$, so assume $(p, p') \neq (0, 0)$. We bound the ratio $h(p + p' - pp') / \frac{1}{2}(h(p) + h(p'))$ from below. First, concavity of h (Item 1.3(iv)) gives

$$\frac{h(p) + h(p')}{2} \leq h\left(\frac{p + p'}{2}\right). \quad (1)$$

Next, on $[0, 0.1]^2$ we have $1 - p' \geq 0.9$, hence

$$p + p' - pp' = p + p'(1 - p) \geq p(1 - p') + p'(1 - p) \geq 0.9(p + p'). \quad (2)$$

Since h is increasing on $[0, 1/2]$ and $0.9(p + p') \leq \frac{p+p'}{2} \cdot 1.8 \leq 0.18 \leq 1/2$, combining Eq. (1) and Eq. (2) yields, with $u := p + p' \in (0, 0.2]$,

$$\frac{h(p + p' - pp')}{\frac{1}{2}(h(p) + h(p'))} \geq \frac{h(0.9(p + p'))}{h(\frac{1}{2}(p + p'))} = \frac{h(0.9u)}{h(0.5u)} =: g(u).$$

A direct calculation shows g is minimized on $(0, 0.2]$ at $u = 0.2$, where

$$g(0.2) = \frac{h(0.18)}{h(0.10)} = 1.450\dots > 1.4.$$

Therefore $h(p + p' - pp') \geq 1.4 \cdot \frac{1}{2}(h(p) + h(p'))$ on the whole domain, as desired. $\square$

Lemma 3.2 (Mixed-probability bound). *For all $p, p' \in [0, 1]$,*

$$h(p + p' - pp') \geq (1 - p) h(p').$$

Proof. Write the OR-probability as a convex combination of the endpoints 1 and p' :

$$p + p' - pp' = p \cdot 1 + (1 - p) p'.$$

Applying concavity of h (Item 1.3(iv)) to this convex combination,

$$h(p \cdot 1 + (1 - p) p') \geq p h(1) + (1 - p) h(p') = (1 - p) h(p'),$$

where the final equality uses $h(1) = 0$ (Definition 1.2). This gives the assertion. $\square$

Remark 3.3 (Relation to the i.i.d.-marginal heuristic). In the special case where A has i.i.d. Bernoulli(p) coordinates, one has $H(A) = h(p) n$ and $H(A \cup B) = h(2p - p^2) n$, so the relevant single-variable quantity is $h(2p - p^2) - 2h(p)$ in disguise (equivalently the ratio $h(2p - p^2)/h(p)$). This is the heuristic placeholder $h(p^2) - 2h(p)$ referenced in the problem statement, recovered from Lemma 3.1 by setting $p = p'$. The two lemmas above are the conditional generalizations actually needed once the coordinates are correlated.

4 Region bounds

We now set up the conditional model of the key lemma and prove the two region bounds that drive it. Let C be a random variable on a finite set S , and for each $c \in S$ fix $p_c \in [0, 1]$. Sample $X \in \{0, 1\}$ by first drawing $c \sim C$ and then setting $\mathbb{P}[X = 1 \mid C = c] = p_c$; let (C', X') be an independent copy of (C, X) . Write $q(c) = \mathbb{P}[C = c]$. We assume throughout this section the marginal cap

$$\mathbb{E}[X] = \mathbb{E}_c[p_c] \leq 0.01. \quad (3)$$

Partition S by the threshold 0.1:

$$\mathcal{C}_0 := \{c \in S : p_c \leq 0.1\}, \quad \mathcal{C}_1 := S \setminus \mathcal{C}_0.$$

We abbreviate the events $\{C \in \mathcal{C}_0\}$, $\{C \in \mathcal{C}_1\}$ by $\mathcal{C}_0$, $\mathcal{C}_1$ when no confusion arises. Markov's inequality applied to Eq. (3) gives the region-probability bound

$$\mathbb{P}[\mathcal{C}_1] = \mathbb{P}[p_C > 0.1] \leq \frac{\mathbb{E}_c[p_c]}{0.1} \leq \frac{0.01}{0.1} = 0.1, \quad \text{so} \quad \mathbb{P}[\mathcal{C}_0] \geq 0.9. \quad (4)$$

Lemma 4.1 (Both-small region). *Under Eq. (3),*

$$\mathbb{P}[\mathcal{C}_0]^2 H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_0) \geq 1.26 \mathbb{P}[\mathcal{C}_0] H(X \mid C \in \mathcal{C}_0).$$

Proof. Let q_0 denote the law of C conditioned on $\mathcal{C}_0$, i.e. $q_0(c) = q(c)/\mathbb{P}[\mathcal{C}_0]$ for $c \in \mathcal{C}_0$. Conditioned on $C \in \mathcal{C}_0$ the bit X is Bernoulli(p_c), so $H(X \mid C \in \mathcal{C}_0) = \mathbb{E}_{c \sim q_0}[h(p_c)]$. We compute

$$\begin{aligned} \mathbb{P}[\mathcal{C}_0] H(X \mid C \in \mathcal{C}_0) &\stackrel{(a)}{=} \mathbb{P}[\mathcal{C}_0] \mathbb{E}_{c \sim q_0}[h(p_c)] \\ &\stackrel{(b)}{=} \mathbb{P}[\mathcal{C}_0] \mathbb{E}_{c, c' \sim q_0} \left[\frac{h(p_c) + h(p_{c'})}{2} \right] \\ &\stackrel{(c)}{\leq} \frac{\mathbb{P}[\mathcal{C}_0]}{1.4} \mathbb{E}_{c, c' \sim q_0} [h(p_c + p_{c'} - p_c p_{c'})] \\ &\stackrel{(d)}{=} \frac{\mathbb{P}[\mathcal{C}_0]}{1.4} H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_0) \\ &\stackrel{(e)}{\leq} \frac{\mathbb{P}[\mathcal{C}_0]^2}{1.26} H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_0), \end{aligned}$$

where step (a) is the conditional Bernoulli identity, step (b) symmetrizes by introducing an independent copy $c' \sim q_0$ (the inner expectation is unchanged since $\mathbb{E}_{c' \sim q_0}[h(p_{c'})] = \mathbb{E}_{c \sim q_0}[h(p_c)]$), step (c) applies Lemma 3.1 pointwise (valid since $p_c, p_{c'} \leq 0.1$ on $\mathcal{C}_0$) and takes the expectation, step (d) identifies $\mathbb{E}_{c, c' \sim q_0}[h(p_c + p_{c'} - p_c p_{c'})] = H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_0)$ because conditioned on $\mathcal{C}_0, \mathcal{C}'_0$ the bits X, X' are independent Bernoulli(p_c), Bernoulli($p_{c'}$) drawn from q_0 , so their OR is Bernoulli($p_c + p_{c'} - p_c p_{c'}$), and step (e) replaces one factor $1/1.4$ by $\mathbb{P}[\mathcal{C}_0]/1.26$, valid since $\mathbb{P}[\mathcal{C}_0] \geq 0.9 = 1.26/1.4$ by Eq. (4) gives $1/1.4 \leq \mathbb{P}[\mathcal{C}_0]/1.26$. Multiplying through by 1.26 yields the assertion. $\square$

Lemma 4.2 (Mixed region). *Under Eq. (3),*

$$2 \mathbb{P}[\mathcal{C}_0, \mathcal{C}'_1] H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_1) \geq 1.62 \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1).$$

Proof. Conditioned on $\{C = c, C' = c'\}$ the OR-bit $X \vee X'$ is Bernoulli($p_c + p_{c'} - p_c p_{c'}$), so its entropy is $h(p_c + p_{c'} - p_c p_{c'})$. We expand the left-hand side over the product law and bound term by term:

$$\begin{aligned} 2 \mathbb{P}[\mathcal{C}_0, \mathcal{C}'_1] H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_1) &= 2 \sum_{c \in \mathcal{C}_0} \sum_{c' \in \mathcal{C}_1} q(c) q(c') h(p_c + p_{c'} - p_c p_{c'}) \\ &\geq 2 \sum_{c \in \mathcal{C}_0} \sum_{c' \in \mathcal{C}_1} q(c) q(c') (1 - p_c) h(p_{c'}) \\ &= 2 \left(\sum_{c \in \mathcal{C}_0} q(c) (1 - p_c) \right) \left(\sum_{c' \in \mathcal{C}_1} q(c') h(p_{c'}) \right) \\ &\geq 2 \left(\sum_{c \in \mathcal{C}_0} q(c) \cdot 0.9 \right) \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1) \\ &= 1.8 \mathbb{P}[\mathcal{C}_0] \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1) \\ &\geq 1.62 \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1), \end{aligned}$$

where the first step is the conditional-entropy expansion just noted, the second step applies Lemma 3.2 with $(p, p') = (p_c, p_{c'})$ inside each summand, the third step factors the double sum, the fourth step uses $1 - p_c \geq 0.9$ for $c \in \mathcal{C}_0$ (since $p_c \leq 0.1$) and recognizes $\sum_{c' \in \mathcal{C}_1} q(c') h(p_{c'}) = \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1)$, the fifth step evaluates $\sum_{c \in \mathcal{C}_0} q(c) = \mathbb{P}[\mathcal{C}_0]$, and the last step uses $\mathbb{P}[\mathcal{C}_0] \geq 0.9$ from Eq. (4) so that $1.8 \mathbb{P}[\mathcal{C}_0] \geq 1.8 \cdot 0.9 = 1.62$. This gives the assertion. $\square$

5 The key per-coordinate lemma

We retain the conditional model of Section 4: C is a random variable on a finite set S , $p_c \in [0, 1]$ for $c \in S$, X is Bernoulli(p_C) given C , and (C', X') is an independent copy of (C, X) . The key lemma amplifies the conditional entropy of a single bit by a constant factor when its marginal is small.

Lemma 5.1 (Key amplification lemma). *Suppose $\mathbb{E}[X] = \mathbb{E}[p_c] \leq 0.01$. Then*

$$H(X \vee X' \mid C, C') \geq 1.26 H(X \mid C).$$

Equivalently, if $\{p_c\}_{c \in S} \subset [0, 1]$ satisfies $\mathbb{E}_c[p_c] \leq 0.01$, then $\mathbb{E}_{c, c'}[h(p_c + p_{c'} - p_c p_{c'})] \geq 1.26 \mathbb{E}_c[h(p_c)]$.

Proof. The strategy is to split the joint conditioning (C, C') into the three disjoint region-events determined by the threshold partition $S = \mathcal{C}_0 \sqcup \mathcal{C}_1$ of Section 4, bound the first two regions by Lemmas 4.1 and 4.2, and discard the third (which is non-negative). Expanding $H(X \vee X' \mid C, C')$ over the four region combinations and grouping the two mixed combinations by symmetry,

$$\begin{aligned} H(X \vee X' \mid C, C') &= \underbrace{\mathbb{P}[\mathcal{C}_0, \mathcal{C}'_0] H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_0)}_{=: T_1} \\ &\quad + \underbrace{2 \mathbb{P}[\mathcal{C}_0, \mathcal{C}'_1] H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_1)}_{=: T_2} \\ &\quad + \underbrace{\mathbb{P}[\mathcal{C}_1, \mathcal{C}'_1] H(X \vee X' \mid \mathcal{C}_1, \mathcal{C}'_1)}_{=: T_3}. \end{aligned} \tag{5}$$

Here T_2 collects both mixed events $\{\mathcal{C}_0, \mathcal{C}'_1\}$ and $\{\mathcal{C}_1, \mathcal{C}'_0\}$, which contribute equally because (C, X) and (C', X') are i.i.d. and the OR is symmetric; this accounts for the factor 2. Using independence of the two copies, $\mathbb{P}[\mathcal{C}_0, \mathcal{C}'_0] = \mathbb{P}[\mathcal{C}_0]^2$ and $\mathbb{P}[\mathcal{C}_0, \mathcal{C}'_1] = \mathbb{P}[\mathcal{C}_0] \mathbb{P}[\mathcal{C}_1]$. We bound each surviving term:

$$\begin{aligned} T_1 &= \mathbb{P}[\mathcal{C}_0]^2 H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_0) \stackrel{(a)}{\geq} 1.26 \mathbb{P}[\mathcal{C}_0] H(X \mid C \in \mathcal{C}_0), \\ T_2 &= 2 \mathbb{P}[\mathcal{C}_0] \mathbb{P}[\mathcal{C}_1] H(X \vee X' \mid \mathcal{C}_0, \mathcal{C}'_1) \stackrel{(b)}{\geq} 1.62 \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1), \\ T_3 &\stackrel{(c)}{\geq} 0, \end{aligned}$$

where step (a) is Lemma 4.1, step (b) is Lemma 4.2 (noting $\mathbb{P}[\mathcal{C}_0, \mathcal{C}'_1] = \mathbb{P}[\mathcal{C}_0] \mathbb{P}[\mathcal{C}_1]$ so the lemma's left-hand side equals T_2), and step (c) holds because entropy is non-negative. Summing the three bounds and using $1.62 \geq 1.26$,

$$\begin{aligned} H(X \vee X' \mid C, C') &\geq 1.26 \mathbb{P}[\mathcal{C}_0] H(X \mid C \in \mathcal{C}_0) + 1.62 \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1) \\ &\geq 1.26 (\mathbb{P}[\mathcal{C}_0] H(X \mid C \in \mathcal{C}_0) + \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1)) \\ &= 1.26 H(X \mid C), \end{aligned}$$

where the final equality is the law of total conditional entropy across the partition $S = \mathcal{C}_0 \sqcup \mathcal{C}_1$: $H(X \mid C) = \mathbb{P}[\mathcal{C}_0] H(X \mid C \in \mathcal{C}_0) + \mathbb{P}[\mathcal{C}_1] H(X \mid C \in \mathcal{C}_1)$. This completes the proof. $\square$

Remark 5.2 (The discarded region). The term T_3 , corresponding to both samples having large conditional marginal, is thrown away. It is non-negative, and regions T_1, T_2 alone already exceed $1.26 H(X \mid C)$, so retaining T_3 would only improve the constant. A tight version of Lemma 5.1 — which would push the Frankl bound to the sharp threshold $\frac{3-\sqrt{5}}{2} \approx 0.38$ — requires analyzing T_3 as well; this was carried out in follow-up work and is not reproduced here.

6 The entropy theorem

We now lift the per-coordinate Lemma 5.1 to the full vector via the chain rule, obtaining the information-theoretic strengthening due to Gilmer [2] (Theorem 1 therein). The theorem holds for an arbitrary distribution on $2^{[n]}$, not only the uniform distribution over a union-closed family.

Theorem 6.1 (Entropy amplification under small marginals). *Let A, B be independent samples from a common distribution over subsets of $[n]$, and suppose $\mathbb{P}[i \in A] \leq 0.01$ for every $i \in [n]$. Then*

$$H(A \cup B) \geq 1.26 H(A).$$

Proof. The strategy is to reveal the coordinates of $A \cup B$ and of A one at a time and to show, coordinate by coordinate, that the conditional entropy of the i -th bit of $A \cup B$ dominates 1.26 times that of the i -th bit of A ; the chain rule then assembles these into the claimed inequality. Fix $i \in [n]$. We claim

$$H((A \cup B)_i \mid (A \cup B)_{<i}) \geq 1.26 H(A_i \mid A_{<i}). \quad (6)$$

To prove Eq. (6), observe first that $(A \cup B)_{<i}$ is a deterministic function of the richer history $(A_{<i}, B_{<i})$; hence

$$\begin{aligned} H((A \cup B)_i \mid (A \cup B)_{<i}) &\stackrel{(a)}{\geq} H((A \cup B)_i \mid A_{<i}, B_{<i}) \\ &\stackrel{(b)}{=} H(A_i \vee B_i \mid A_{<i}, B_{<i}) \\ &\stackrel{(c)}{\geq} 1.26 H(A_i \mid A_{<i}), \end{aligned} \quad (7)$$

where step (a) is Lemma 2.1 applied with $X = (A \cup B)_i$, $Y = (A_{<i}, B_{<i})$, and $f(Y) = (A \cup B)_{<i}$; step (b) rewrites the i -th union bit as the OR $A_i \vee B_i$; and step (c) is Lemma 5.1 with $C = A_{<i}$, $C' = B_{<i}$, $X = A_i$, $X' = B_i$. The hypotheses of Lemma 5.1 are met: $A_{<i}$ and $B_{<i}$ are i.i.d. over a common finite set, A_i is conditionally Bernoulli given $A_{<i}$ (with parameter $p_{A_{<i}} = \mathbb{P}[A_i = 1 \mid A_{<i}]$), B_i given $B_{<i}$ is the independent copy, and the marginal cap holds because $\mathbb{E}[A_i] = \mathbb{P}[i \in A] \leq 0.01$. This establishes Eq. (6).

Summing Eq. (6) over $i \in [n]$ and applying the chain rule Item 1.3(ii) to each side,

$$H(A \cup B) = \sum_{i=1}^n H((A \cup B)_i \mid (A \cup B)_{<i}) \geq 1.26 \sum_{i=1}^n H(A_i \mid A_{<i}) = 1.26 H(A).$$

This completes the proof. $\square$

7 The union-closed corollary

We finally specialize Theorem 6.1 to the uniform distribution over a union-closed family, recovering Gilmer's constant lower bound for the union-closed sets conjecture [2] (Theorem 2 therein).

Theorem 7.1 (Constant lower bound for union-closed families). *Let $\mathcal{F} \subseteq 2^{[n]}$ be a union-closed family with $|\mathcal{F}| \geq 2$. Then there exists an element $i \in [n]$ contained in at least a $c = 0.01$ fraction of the sets of $\mathcal{F}$; that is,*

$$|\{A \in \mathcal{F} : i \in A\}| \geq 0.01 |\mathcal{F}|.$$

Proof. The strategy is to assume every element has frequency below 0.01 and derive a contradiction with Theorem 6.1. Let A, B be independent uniform samples from $\mathcal{F}$, and set $p_i := \mathbb{P}[i \in A] = |\{A \in \mathcal{F} : i \in A\}|/|\mathcal{F}|$. Suppose, for contradiction, that $p_i < 0.01$ for every $i \in [n]$; in particular $\mathbb{P}[i \in A] \leq 0.01$ for all i , so Theorem 6.1 applies and gives

$$H(A \cup B) \geq 1.26 H(A). \quad (8)$$

On the other hand, union-closure forces $A \cup B \in \mathcal{F}$ almost surely, so $A \cup B$ is supported on $\mathcal{F}$; the maximum-entropy bound Item 1.3(i) and the fact that A is uniform on $\mathcal{F}$ give

$$H(A \cup B) \stackrel{(a)}{\leq} \log |\mathcal{F}| \stackrel{(b)}{=} H(A), \quad (9)$$

where step (a) applies Item 1.3(i) to the $\mathcal{F}$ -supported variable $A \cup B$, and step (b) uses that A is uniform on $\mathcal{F}$, for which Item 1.3(i) holds with equality $H(A) = \log |\mathcal{F}|$. Since $|\mathcal{F}| \geq 2$, the uniform distribution on $\mathcal{F}$ is non-degenerate, so

$$H(A) = \log |\mathcal{F}| \geq \log 2 = 1 > 0.$$

Chaining Eq. (9) and Eq. (8),

$$H(A) \stackrel{(a)}{\geq} H(A \cup B) \stackrel{(b)}{\geq} 1.26 H(A) \stackrel{(c)}{>} H(A),$$

where step (a) is Eq. (9), step (b) is Eq. (8), and step (c) is strict because $H(A) > 0$. The chain $H(A) > H(A)$ is a contradiction. Hence some $p_i \geq 0.01$, i.e. some element i lies in at least a 0.01 fraction of the sets of $\mathcal{F}$. This completes the proof. $\square$

Remark 7.2 (On the constant). The constant 0.01 is not optimized; it is the value carried through the unoptimized region constants of Lemmas 4.1 and 4.2. The sharp threshold for this method is $\frac{3-\sqrt{5}}{2} \approx 0.38$, attained by later refinements of Lemma 5.1; Frankl's conjecture itself asks for the constant $\frac{1}{2}$, which remains open and provably out of reach of the present argument since $\frac{3-\sqrt{5}}{2} < \frac{1}{2}$.

References

- [1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.
- [2] Justin Gilmer. A constant lower bound for the union-closed sets conjecture, 2022. arXiv:2211.09055.